THE MASTERMIND COACHING CLASSES

CBSE / RBSE · Nursery – Class XII · Road No. 5, VKIA, Jaipur · 9887258679

CLASS IX — MATHEMATICS · FULL SYLLABUS EXAMINATION

Session 2026-27 | CBSE / RBSE Pattern

Class: 9	Subject: Maths	Set 05	Time: 3 Hours	Max. Marks: 80

General Instructions:

1. This question paper contains 38 questions and is divided into 5 sections — A, B, C, D and E.
2. In Section A, Q. 1–18 are multiple-choice questions and Q. 19–20 are Assertion–Reason based questions of 1 mark each.
3. In Section B, Q. 21–25 are very short answer (VSA) type questions carrying 2 marks each.
4. In Section C, Q. 26–31 are short answer (SA) type questions carrying 3 marks each.
5. In Section D, Q. 32–35 are long answer (LA) type questions carrying 5 marks each.
6. In Section E, Q. 36–38 are case-study based questions carrying 4 marks each with sub-parts of 1, 1 and 2 marks respectively.
7. All questions are compulsory. However, an internal choice is provided in 1 question of Section B, 2 questions of Section C and 2 questions of Section D. An internal choice is provided in all the 2-mark sub-parts of Section E.
8. Draw neat and clean figures wherever required. Take $\pi = 22/7$ wherever required, if not stated. Use of calculators is not allowed.
9. Write the serial number of each question before attempting it.

SECTION A — Objective Type — 20 questions · 20 marks

Q1–Q18 are multiple-choice questions (1 mark each). Q19–Q20 are Assertion (A) and Reason (R) questions: select the correct option from (a), (b), (c), (d) given below each.

1. Which of the following points lies on neither axis? **1**

(a) (−4, 0) (b) (3, −2) (c) (0, 5) (d) (0, 0)

2. The distance between the points (−2, 0) and (2, 0) is: **1**

(a) 2 units (b) 4 units (c) 0 units (d) 8 units

3. If the point P(a, 3) is at a distance of 5 units from the point (0, −1), then the value of a is: **1**

(a) ±4 (b) 3 only (c) ±5 (d) ±3

4. If $p(x) = 4x^2 - 3x + 7$, then p(0) is: **1**

(a) 0 (b) 8 (c) 7 (d) −3

5. The degree of the polynomial $(x + 1)(x - 1)$ is: **1**

(a) 0 (b) 1 (c) 2 (d) 3

6. The area of a rectangular garden is $(x^2 - 9)$ m² and its breadth is $(x - 3)$ m. Its length is: **1**

(a) $(x^2 - 6)$ m (b) $(x - 3)$ m (c) $(x + 3)$ m (d) $(x + 9)$ m

7. Which of the following has a non-terminating repeating decimal expansion? **1**

(a) $\frac{7}{25}$ (b) $\frac{5}{6}$ (c) $\frac{9}{40}$ (d) $\frac{3}{16}$

8. The irrational number $\sqrt{3}$ lies between: **1**

(a) 1 and 2 (b) 2 and 3 (c) 3 and 4 (d) 0 and 1

9. The value of $|3 - 7| + |-2|$ is: **1**

(a) −6 (b) −2 (c) 6 (d) 2

10. The expression $x^2 - 6x + 9$ can be written as: **1**

(a) $(x + 3)^2$ (b) $(x - 9)^2$ (c) $(x - 3)(x + 3)$ (d) $(x - 3)^2$

11. If $a + b + c = 10$ and $ab + bc + ca = 31$, then $a^2 + b^2 + c^2$ is: **1**

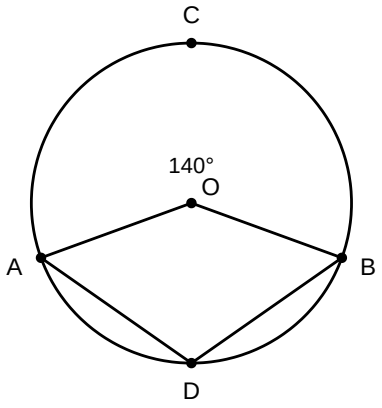
(a) 100 (b) 38 (c) 69 (d) 41

12. In a circle with centre O, chords AB and CD are equal. If the perpendicular distance of AB from O is 7 cm, the perpendicular distance of CD from O is: **1**

(a) cannot be found (b) 3.5 cm

(c) 7 cm (d) 14 cm

13. In the figure, O is the centre of the circle and ∠AOB = 140°. If D is a point on the minor arc AB, then ∠ADB is: **1**



(a) 220° (b) 110° (c) 70° (d) 140°

14. The area of an equilateral triangle of side 8 cm is: **1**

(a) $64\sqrt{3}$ cm² (b) $8\sqrt{3}$ cm² (c) $16\sqrt{3}$ cm² (d) $32\sqrt{3}$ cm²

15. The area of a triangle with sides 9 cm, 10 cm and 17 cm is: **1**

(a) 36 cm² (b) 45 cm² (c) 72 cm² (d) 90 cm²

16. Two dice are thrown together. The probability that the sum of the numbers is 12 is: **1**

(a) $\frac{1}{6}$ (b) $\frac{1}{36}$ (c) $\frac{1}{18}$ (d) $\frac{1}{12}$

17. A letter is chosen at random from the word STATISTICS. The probability that it is T is: **1**

(a) $\frac{2}{5}$ (b) $\frac{3}{10}$ (c) $\frac{1}{5}$ (d) $\frac{1}{10}$

18. If the n-th term of a sequence is $t_n = n^2 + 1$, then t_9 is: **1**

(a) 19 (b) 10 (c) 81 (d) 82

19. **Assertion (A):** The point (−3, 0) lies on the x-axis. **1**
Reason (R): Every point on the x-axis has ordinate 0.

(a) Both A and R are true, and R is the correct explanation of A.

(b) Both A and R are true, but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

20. **Assertion (A):** $(a + b)^2 = a^2 + b^2$ for all real numbers a and b. **1**
Reason (R): $(a + b)^2 = a^2 + 2ab + b^2$.

(a) Both A and R are true, and R is the correct explanation of A.

(b) Both A and R are true, but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

SECTION B — Very Short Answer (2 marks each) — 5 questions · 10 marks

Internal choice is provided in one question.

21. A courier company charges ₹40 for the first 500 g and ₹15 for every additional 100 g (or part thereof). Write the charge for n additional units of 100 g and find the charge for a parcel of 1.2 kg. **2**

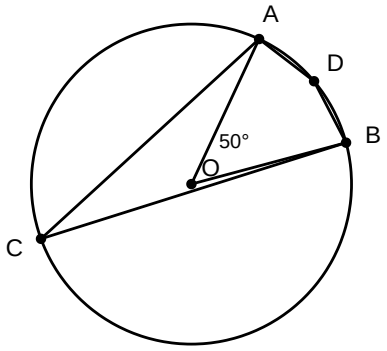
22. Without performing division, show that $\frac{49}{2^3 \times 5^4 \times 7}$ is a terminating decimal and state the number of decimal places. **2**

OR

Find two rational numbers between $\frac{5}{8}$ and $\frac{3}{4}$.

23. Factorise: $49x^2 + 84xy + 36y^2$. **2**

24. In the figure, O is the centre of the circle and $\angle AOB = 50°$. Find $\angle ACB$ and $\angle ADB$. **2**



25. Find the second, third and fourth terms of the sequence defined by $t_1 = 3$ and $t_n = 2t_{n-1} + n$ for $n \geq 2$. **2**

SECTION C — Short Answer (3 marks each) — 6 questions · 18 marks

Internal choice is provided in two questions.

26. Express $0.363636\ldots + 0.272727\ldots$ as a fraction in the form p/q. **3**

27. A coin was tossed 80 times and the number of heads was recorded: **3**

Heads	3	2	1	0
Frequency	10	28	30	12

Find the probability of getting (i) at least 2 heads, (ii) exactly 1 head, (iii) no head.

OR

A die was thrown 60 times with the following results:

Outcome	1	2	3	4	5	6
Frequency	8	12	10	9	11	10

Find the probability of getting (i) an even number, (ii) a number greater than 4, (iii) the number 1.

28. Find the distance between the points A(−6, 7) and B(−1, −5), and the coordinates of the midpoint of AB. **3**

29. Factorise $x^3 - 6x^2 + 12x - 8$ and hence find its value for $x = 12$. **3**

30. Prove that the perpendicular drawn from the centre of a circle to a chord bisects the chord. **3**

31. The sides of a triangular park are in the ratio 7 : 24 : 25 and its perimeter is 168 m. Find its area using Heron's formula. **3**

OR

Find the area of a triangle with sides 9 cm, 40 cm and 41 cm using Heron's formula.

SECTION D — Long Answer (5 marks each) — 4 questions · 20 marks

Internal choice is provided in two questions.

32. Riya buys x pencils at ₹5 each and y erasers at ₹3 each and spends exactly ₹30. **5**

(a) Write this as a linear equation in x and y, in the form ax + by + c = 0, and write a, b, c. (1 mark)

(b) Complete a table of three solutions (x, y). (1 mark)

(c) Draw the graph of the equation on a graph paper. (2 marks)

(d) Use the graph to find the number of erasers she buys if she buys 3 pencils. (1 mark)

33. (a) How many odd numbers lie between 100 and 400? (b) Find their sum. (c) Is 305 a term of this list? Give a reason. (2 + 2 + 1 marks) **5**

34. The lifetimes of 1000 electric bulbs were recorded: **5**

Lifetime (hours)	0 – 300	300 – 600	600 – 900	900 – 1200
Number of bulbs	40	160	450	350

A bulb is chosen at random. Find the probability that its lifetime is (i) less than 300 hours, (ii) between 300 and 900 hours, (iii) more than 900 hours.

OR

A survey of 200 families recorded the number of children in each family:

Children	0	1	2	3
Families	22	90	70	18

A family is chosen at random. Find the probability that it has (i) exactly 2 children, (ii) at most 1 child, (iii) more than 2 children.

35. The length of a rectangular park is 3 m more than its breadth and its area is 130 m². Form a quadratic expression, factorise it to find the dimensions, and find the cost of fencing the park at ₹40 per metre. **5**

OR

A mosaic is made of 2 big square tiles of side x, 9 long strips of size x × 1 and 4 small unit squares, so its area is $(2x^2 + 9x + 4)$. Factorise the expression to find the dimensions of the rectangle formed. Find its perimeter and area when x = 5.

SECTION E — Case-Study Based Questions (4 marks each) — 3 questions · 12 marks

Read each case carefully and answer the questions that follow. Internal choice is provided in every 2-mark sub-part.

36. A triangular plot of land has sides 25 m, 29 m and 36 m.

(i) Find the semi-perimeter of the plot. 1

(ii) Find the area of the plot using Heron's formula. 1

(iii) The plot is to be levelled at ₹5 per m². Find the total cost. 2

OR

(iii) Find the altitude of the plot on its longest side.

37. In a school sports day, 120 entries were received for four events:

Event	Races	Jumps	Throws	Relays
Entries	36	30	24	30

One entry form is picked at random.

(i) Find the probability that it is for a race. 1

(ii) Find the probability that it is not for a throw. 1

(iii) Find the probability that it is for a jump or a relay. 2

OR

(iii) If 20 more entries are received, all for throws, find the new probability that a randomly picked entry is for a throw.

38. Kabir saves money in a jar. He starts with ₹200 and adds ₹40 every week. The amount in the jar after w weeks is M = 200 + 40w.

(i) How much money is there in the jar after 6 weeks? 1

(ii) What does the number 40 in the equation represent? 1

(iii) After how many weeks will the jar have ₹800? 2

OR

(iii) Find the amount after 12 weeks and its increase over the amount after 4 weeks.